EXERCISE 5

Building a Matrix from a Complex Pair of Eigenvalues and Eigenvectors

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Inverse Problem • A = Q J Q^{-1} • Three Matrices

STATEMENT

Show how to construct a matrix with a complex conjugate pair of eigenvalues $\lambda_\pm = \rho \pm i\omega$ and corresponding eigenvectors $\boldsymbol{\eta}_\pm = \mathbf{v}_1 \pm i\mathbf{v}_2$. Hence find the matrices with the following eigenvalues and eigenvectors:

- (i) $\lambda_\pm = 3 \pm 3i,\ \boldsymbol{\eta}_\pm = (2 \pm i,\ 1 \mp i)$;
- (ii) $\lambda_\pm = \pm 3i,\ \boldsymbol{\eta}_\pm = (\pm i,\ 3 \pm 2i)$;
- (iii) $\lambda_\pm = -2 \pm i,\ \boldsymbol{\eta}_\pm = (1 \pm i,\ 1 \mp i)$.

PHASE 1: SETUP AND HYPOTHESES

We look for a real matrix A (2×2) with $A\boldsymbol{\eta}_+ = \lambda_+ \boldsymbol{\eta}_+$ where $\boldsymbol{\eta}_+ = \mathbf{v}_1 + i\mathbf{v}_2$ and $\lambda_+ = \rho + i\omega$, with $\mathbf{v}_1, \mathbf{v}_2$ real and independent and $\omega \neq 0$. Because A is real, $A\bar{\boldsymbol{\eta}}_+ = \bar{\lambda}_+ \bar{\boldsymbol{\eta}}_+$ follows automatically. (Note on the statement: the sheet writes $\boldsymbol{\eta}_\pm = \mathbf{v}_1 \pm i\mathbf{v}_2$ for the eigenvectors of $\lambda_\pm = \rho \pm i\omega$, i.e. $\mathbf{v}_1 = \mathbf{u}$, $\mathbf{v}_2 = \mathbf{w}$ of Topic 4.)

PHASE 2: FORMULATION: THE CONSTRUCTION

Expand $A(\mathbf{v}_1 + i\mathbf{v}_2) = (\rho + i\omega)(\mathbf{v}_1 + i\mathbf{v}_2)$ and separate real and imaginary parts (both $A\mathbf{v}_j$ are real vectors, $i^2 = -1$):

$$A\mathbf{v}_1 + iA\mathbf{v}_2 = (\rho\mathbf{v}_1 - \omega\mathbf{v}_2) + i(\omega\mathbf{v}_1 + \rho\mathbf{v}_2) \implies A\mathbf{v}_1 = \rho\mathbf{v}_1 - \omega\mathbf{v}_2, \quad A\mathbf{v}_2 = \omega\mathbf{v}_1 + \rho\mathbf{v}_2$$

Collect the two vector equations in one matrix equation, with $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$ (columns): the first column of QJ is $\rho\mathbf{v}_1 - \omega\mathbf{v}_2$ for $J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$ and the second is $\omega\mathbf{v}_1 + \rho\mathbf{v}_2$. Hence $AQ = QJ$, and since $\mathbf{v}_1, \mathbf{v}_2$ are independent (Exercise 4) Q is invertible:

$$\boxed{A = QJQ^{-1}, \qquad Q = [\mathbf{v}_1 \mid \mathbf{v}_2], \qquad J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}}$$

This A is real (a product of real matrices), has the eigenvalues of J, $\rho \pm i\omega$ (similar matrices have the same characteristic polynomial), and $A\boldsymbol{\eta}_+ = \lambda_+ \boldsymbol{\eta}_+$ by construction.

PHASE 3: DERIVATION OF THE THREE MATRICES

(i) $\lambda_\pm = 3 \pm 3i$, $\boldsymbol{\eta}_\pm = (2 \pm i,\ 1 \mp i)$

$\boldsymbol{\eta}_+ = \begin{pmatrix} 2+i \\ 1-i \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + i \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, so $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $\rho = 3, \omega = 3$.

$$Q = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 3 & 3 \\ -3 & 3 \end{pmatrix}, \quad \det Q = -3, \quad Q^{-1} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{pmatrix}$$

$$\begin{aligned} (QJ)_{11} &= 2 \cdot 3 + 1 \cdot (-3) = 3 \\ (QJ)_{12} &= 2 \cdot 3 + 1 \cdot 3 = 9 \\ (QJ)_{21} &= 1 \cdot 3 + (-1) \cdot (-3) = 6 \\ (QJ)_{22} &= 1 \cdot 3 + (-1) \cdot 3 = 0 \end{aligned}$$

$$\begin{aligned} (A)_{11} &= 3 \cdot \frac{1}{3} + 9 \cdot \frac{1}{3} = 4 \\ (A)_{12} &= 3 \cdot \frac{1}{3} + 9 \cdot (-\frac{2}{3}) = -5 \\ (A)_{21} &= 6 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} = 2 \\ (A)_{22} &= 6 \cdot \frac{1}{3} + 0 \cdot (-\frac{2}{3}) = 2 \end{aligned}$$

$$A = QJQ^{-1} = \begin{pmatrix} 4 & -5 \\ 2 & 2 \end{pmatrix}$$

Verification

$$A\boldsymbol{\eta}_+ = \begin{pmatrix} 4 & -5 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 2+i \\ 1-i \end{pmatrix} = \begin{pmatrix} 4\,(2+i) + (-5)\,(1-i) = 3+9i \\ 2\,(2+i) + 2\,(1-i) = 6 \end{pmatrix} = (3+3i) \begin{pmatrix} 2+i \\ 1-i \end{pmatrix}$$

Characteristic polynomial: $\lambda^2 - (6)\lambda + (18) = 0$ has $\operatorname{tr} A = 6 = 2\rho$ and $\det A = 18 = \rho^2 + \omega^2$, hence $\lambda = 3 \pm i\sqrt{9}$, i.e. $\rho \pm i\omega$ with $\rho = 3, \omega = 3$.

(ii) $\lambda_\pm = \pm 3i,\ \boldsymbol{\eta}_\pm = (\pm i,\ 3 \pm 2i)$

$\boldsymbol{\eta}_+ = \begin{pmatrix} i \\ 3+2i \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix} + i \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, so $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 3 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \rho = 0, \omega = 3.$

$$Q = \begin{pmatrix} 0 & 1 \\ 3 & 2 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}, \quad \det Q = -3, \quad Q^{-1} = \begin{pmatrix} -\frac{2}{3} & \frac{1}{3} \\ 1 & 0 \end{pmatrix}$$

$$\begin{aligned} (QJ)_{11} &= 0 \cdot 0 + 1 \cdot (-3) = -3 \\ (QJ)_{12} &= 0 \cdot 3 + 1 \cdot 0 = 0 \\ (QJ)_{21} &= 3 \cdot 0 + 2 \cdot (-3) = -6 \\ (QJ)_{22} &= 3 \cdot 3 + 2 \cdot 0 = 9 \end{aligned}$$

$$\begin{aligned} (A)_{11} &= (-3) \cdot (-\frac{2}{3}) + 0 \cdot 1 = 2 \\ (A)_{12} &= (-3) \cdot \frac{1}{3} + 0 \cdot 0 = -1 \\ (A)_{21} &= (-6) \cdot (-\frac{2}{3}) + 9 \cdot 1 = 13 \\ (A)_{22} &= (-6) \cdot \frac{1}{3} + 9 \cdot 0 = -2 \end{aligned}$$

$$A = QJQ^{-1} = \begin{pmatrix} 2 & -1 \\ 13 & -2 \end{pmatrix}$$

Verification

$$A\boldsymbol{\eta}_+ = \begin{pmatrix} 2 & -1 \\ 13 & -2 \end{pmatrix}\begin{pmatrix} i \\ 3+2i \end{pmatrix} = \begin{pmatrix} 2\,(i) + (-1)\,(3+2i) = -3 \\ 13\,(i) + (-2)\,(3+2i) = -6+9i \end{pmatrix} = (3i)\,\begin{pmatrix} i \\ 3+2i \end{pmatrix}$$

Characteristic polynomial: $\lambda^2 - (0)\lambda + (9) = 0$ has $\operatorname{tr} A = 0 = 2\rho$ and $\det A = 9 = \rho^2 + \omega^2$, hence $\lambda = 0 \pm i\sqrt{9}$, i.e. $\rho \pm i\omega$ with $\rho = 0, \omega = 3$.

(iii) $\lambda_\pm = -2 \pm i,\ \boldsymbol{\eta}_\pm = (1 \pm i,\ 1 \mp i)$

$\boldsymbol{\eta}_+ = \begin{pmatrix} 1+i \\ 1-i \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + i\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, so $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \rho = -2, \omega = 1.$

$$Q = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} -2 & 1 \\ -1 & -2 \end{pmatrix}, \quad \det Q = -2, \quad Q^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\begin{aligned} (QJ)_{11} &= 1\cdot(-2) + 1\cdot(-1) = -3 \\ (QJ)_{12} &= 1\cdot 1 + 1\cdot(-2) = -1 \\ (QJ)_{21} &= 1\cdot(-2) + (-1)\cdot(-1) = -1 \\ (QJ)_{22} &= 1\cdot 1 + (-1)\cdot(-2) = 3 \end{aligned}$$

$$\begin{aligned} (A)_{11} &= (-3)\cdot\frac{1}{2} + (-1)\cdot\frac{1}{2} = -2 \\ (A)_{12} &= (-3)\cdot\frac{1}{2} + (-1)\cdot(-\frac{1}{2}) = -1 \\ (A)_{21} &= (-1)\cdot\frac{1}{2} + 3\cdot\frac{1}{2} = 1 \\ (A)_{22} &= (-1)\cdot\frac{1}{2} + 3\cdot(-\frac{1}{2}) = -2 \end{aligned}$$

$$A = QJQ^{-1} = \begin{pmatrix} -2 & -1 \\ 1 & -2 \end{pmatrix}$$

Verification

$$A\boldsymbol{\eta}_+ = \begin{pmatrix} -2 & -1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 1+i \\ 1-i \end{pmatrix} = \begin{pmatrix} (-2)(1+i) + (-1)(1-i) = -3-i \\ 1(1+i) + (-2)(1-i) = -1+3i \end{pmatrix} = (-2+i) \begin{pmatrix} 1+i \\ 1-i \end{pmatrix}$$

Characteristic polynomial: $\lambda^2 - (-4)\lambda + (5) = 0$ has $\operatorname{tr} A = -4 = 2\rho$ and $\det A = 5 = \rho^2 + \omega^2$, hence $\lambda = -2 \pm i\sqrt{1}$, i.e. $\rho \pm i\omega$ with $\rho = -2, \omega = 1$.

PHASE 4: INTERPRETATION AND CHECKS

- The matrix is **not unique** if only the eigenvalues are prescribed, but it is unique once $\boldsymbol{\eta}_+$ is given (up to the complex scalar multiple that every eigenvector admits: rescaling $\boldsymbol{\eta}_+$ by $c = c_r + ic_i$ changes $(\mathbf{v}_1, \mathbf{v}_2)$ and leaves A unchanged, since $A\bar{\boldsymbol{\eta}}_+$, $A\boldsymbol{\eta}_+$ still determine A on the basis $\{\boldsymbol{\eta}_+, \bar{\boldsymbol{\eta}}_+\}$ of $\mathbb{C}^2$).
- For the three items $\operatorname{tr} A = 6, 0, -4$ and $\det A = 18, 9, 5$ respectively, which are 2ρ and $\rho^2 + \omega^2$ as the construction predicts.
- (ii) has $\rho = 0$: a centre. $A = \begin{pmatrix} 2 & -1 \\ 13 & -2 \end{pmatrix}$ has trace zero and determinant $9 = \omega^2$.
- This is the inverse of Exercise 3: from A to (Q, J) there, from (Q, J) to A here.